

EXPLOITING STEREO SPATIAL PROPERTIES WITH RESNET-CONFORMERS FOR ROBUST EVENT DETECTION AND LOCALIZATION

Technical Report

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ABSTRACT

This technical report presents our submission for Task 3, Track A of the DCASE 2025 Challenge: Stereo Sound Event Localization and Detection (SELD) in regular video content. Our system focuses exclusively on the audio-only track, leveraging stereo audio inputs for precise SELD. At the heart of our framework lies a ResNet-Conformer architecture augmented with pre-trained ONE-PEACE embeddings. This powerful combination processes a rich set of spatial and spectral features, including Mel spectrograms, Interaural Phase Differences (IPDs), Interaural Level Differences (ILDs), and Generalized Cross-Correlation with Phase Transform (GCC-PHAT). To optimize performance across multiple SELD tasks - namely Direction of Arrival (DoA) estimation, sound event detection (SED), and sound distance estimation (SDE) - we employ a modular training strategy: separate modules are trained for DoA estimation and SDE, with their outputs fused through a joint prediction scheme. The ONE-PEACE embeddings are integrated alongside the ResNet-Conformer outputs and jointly processed to enhance downstream task performance further. We evaluate our system on the DCASE 2025 Task 3 dev-test set, derived from the Sony-TAu Realistic Spatial Soundscapes 2023 (STARSS23) dataset. Our method achieves an F-score of 48.2%, representing a substantial 25.42% improvement over the official baseline, thereby demonstrating the effectiveness of our approach.

Index Terms— ResNet, Conformer, ONE-PEACE, Ensemble Learning

1. INTRODUCTION

Sound Event Localization and Detection (SELD) is a joint task that involves identifying the temporal activity of multiple sound event classes and estimating their spatial trajectories when active. It has received considerable attention within the Detection and Classification of Acoustic Scenes and Events (DCASE) community due to its broad applicability across domains such as surveillance, autonomous robotics, conferencing systems, smart environments, and scene understanding - especially in scenarios involving multichannel or spatial audio capture. The DCASE challenge series initially focused on Sound Event Detection (SED) but has progressively evolved to include spatial localization, reflecting a natural advancement towards more comprehensive acoustic scene analysis. The

2023 and 2024 editions emphasized recordings from ambisonic microphones, which provide full 360° spatial context. In contrast, the 2025 DCASE Challenge shifts focus to SELD using stereo (two-channel) audio, a more accessible format commonly found in consumer video recordings.

This year's task targets systems designed to process stereo audio extracted from everyday video content. Stereo signals inherently contain spatial cues that listeners rely on to infer the positions of sound sources. With the growing prevalence of generative AI technologies capable of producing stereo audio, analyzing such content has become increasingly important - not only for perceptual accuracy but also for downstream cognitive applications such as media summarization, spatial scene understanding, and multimodal content analysis.

The 2025 SELD task is divided into two tracks:

- Track A: Audio-only inference
- Track B: Audiovisual inference

This technical report focuses on Track A, where participants are required to detect predefined sound event classes, track their temporal activity, and estimate their Direction of Arrival (DoA) or spatial positions. The resulting spatio-temporal representation of the acoustic scene enables a wide array of intelligent applications, including environmental inference, self-localization, sound source tracking (even under visual occlusion), and audio-driven automation or monitoring.

Unlike previous challenges that utilized four-channel input formats such as First-Order Ambisonics (FOA), the stereo SELD task presents new challenges due to angular ambiguities inherent in stereo recordings - particularly along the top-bottom and front-back axes. As a result, DoA estimation in this task is constrained to azimuth angles (i.e., left-right localization). The task also continues to emphasize distance estimation, highlighting the need for innovative solutions under limited spatial information.

In recent years, ResNet-Conformer architectures have consistently demonstrated strong performance on SELD tasks, frequently topping the DCASE leaderboards [1]. Continuing this trend, we adopt a similar backbone for stereo SELD. However, we introduce a novel combination of features specifically designed to exploit inter-channel relationships present in stereo audio. We first evaluate these stereo features against the official baseline and then integrate them into our full framework - ReCoOP, which combines the ResNet-Conformer architecture with ONE-PEACE embeddings.

This technical report first introduces the stereo features, many of which are inspired by widely used stereo compression techniques that capture inter-channel dependencies. We then detail our deep

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learning framework, ReCoOP, and evaluate its performance on the DCASE 2025 Task 3 development dataset.

The remainder of this report is organized as follows: Section 2 introduces the dataset used in this study. Section 3 describes the feature extraction methods, including both spatial and contextual audio features. Section 4 outlines the data augmentation strategies applied during training. Section 5 details the proposed ReCoOP deep learning framework. Section 6 presents our experimental setup, and Section 7 discusses the results and key observations. Finally, Section 8 concludes the report with a summary of our findings and potential future directions.

2. DATASET

We use the DCASE2025 Task 3 Stereo SELD Dataset, derived from the STARSS23 dataset [2], which comprises 5-second stereo audio clips simulating realistic audiovisual scenes in indoor environments. The audio is converted from FOA to stereo using mid-side emulation [3], and annotated with sound event class, azimuth (in the left-right plane), distance, and onscreen/offscreen status. The dataset contains 30,000 development clips (41.7 hrs) and 10,000 evaluation clips (13.9 hrs), covering 13 target sound classes with overlapping events and real-world acoustic variability.

3. FEATURE EXTRACTION

This section outlines the acoustic feature extraction strategies used for input representation. We adopt a dual-pronged approach: a physics-inspired spatial feature extractor for direction-aware log-mel spectrograms, and a transformer-based semantic embedding using ONE-PEACE for enhanced global context representation.

3.1. Audio Features Extraction

For each input stereo waveform, we extract a compact yet expressive set of spatial audio features. A Short-Time Fourier Transform (STFT) with a Hann window of length 960 and hop size of 480 samples is applied.

To model spatial cues, the complex STFT outputs of the left and right channels are first projected onto the mel scale using a non-trainable mel filterbank with 64 bins. Three inter-channel directional features are computed from the mel spectrograms [4]:

- Interaural Level Difference (ILD): the logarithmic ratio of spectral energies across channels,
- Interaural Phase Difference (IPD): decomposed into sin and cosine between channels.

Additionally, Generalized Cross-Correlation (GCC) features are derived by computing the phase-only inverse FFT of the complex cross-spectrum, which captures inter-channel time-delay information critical for direction-of-arrival estimation.

The final feature tensor is constructed by concatenating the log-mel spectrograms with ILD, IPD (both sine and cosine components), and GCC features, resulting in a multi-channel representation of shape (6, 251, 64) for each audio clip, where 6 is the number of feature channels, 251 is the number of time bins and 64 is the number of mel bins.

3.2. ONE-PEACE Feature Extraction

To supplement the local spectral features with global semantic cues, we extract contextual embeddings from the ONE-PEACE multi-

modal transformer model [5]. The raw waveform is segmented into overlapping 100 ms chunks with a hop size equal to the segment length. Each segment is processed by the ONE-PEACE audio front-end, which applies tokenization, attention-based encoding, and feature projection.

During inference, each segment is individually passed through the ONE-PEACE model to obtain a fixed-length vector embedding. All segment-wise embeddings are stacked temporally to form a dense contextual representation of the entire audio file, shaped as (1, 50, 1536) where 1 is the batch size, 50 is the number of segments, and 1536 is the embedding dimension.

These embeddings are designed to capture long-range dependencies and event semantics, making them suitable for hybrid architectures where local spectral features are fused with global context for downstream tasks such as sound event detection or localization.

4. DATA AUGMENTATION

To improve the robustness and generalization of our models, we utilized the publicly released DCASE 2024 simulated dataset [6], which contains synthetic recordings generated using the TAU-SRIR database and isolated sound samples. We downmixed this dataset to produce 30,000 stereo audio clips consistent with the format of the DCASE2025 Task 3 setup. Additionally, we applied Audio Channel Swapping (ACS) [7] as a simple but effective augmentation strategy to simulate spatial variance during training [8].

5. RESNET-CONFORMER-ONEPEACE (RECOOP) DEEP LEARNING FRAMEWORK

To address the challenges of spatial sound understanding in realistic stereo recordings, we propose ResNet-Conformer-OnePeace (ReCoOP), a deep learning framework designed for joint sound event detection (SED), direction-of-arrival (DOA) estimation, and source distance estimation. ReCoOP builds upon a shared ResNet-Conformer backbone that efficiently captures spatial and temporal patterns in the input features. The input to the model comprises multi-channel spectro-temporal representations that encode interaural level and phase differences, directional cues, and log-mel spectrograms.

At its core, ReCoOP employs a modified ResNet-18 encoder [9] without final pooling layers to extract localized features from input frames. The convolutional outputs are flattened across time and projected into a fixed embedding dimension. A stack of eight Conformer blocks [10] processes this sequence to model long-range temporal dependencies, combining self-attention and convolutional operations to retain both global context and local detail. Task-specific output heads are then used to produce the required predictions: multi-label sigmoid activation for SED, a tanh-activated head for estimating azimuthal DOA components, and a ReLU-activated head for predicting class-wise source distances [1].

To enhance semantic understanding, one variant of the framework integrates contextual embeddings from the pre-trained ONE-PEACE model. These embeddings, extracted from the raw audio waveform, are temporally aligned with the conformer outputs and fused through concatenation, followed by a secondary projection and refinement using two additional Conformer blocks. This ONE-PEACE-enhanced architecture enables the model to incorporate high-level audiovisual semantics, which are particularly useful for localization in cluttered or ambiguous acoustic scenes.

Joint prediction in ReCoOP is performed through an ensemble of two model variants. The SED output is computed by averaging the predictions from both the SED-SDE and SED-DOA-OnePeace models, thereby balancing spatial and semantic detection capabilities. The DOA output is taken from the SED-DOA-OnePeace model, which is better equipped to model directional information due to its access to global context [1]. The distance estimation is provided exclusively by the SED-SDE model, which is explicitly optimized for this regression task. This ensemble mechanism enables ReCoOP to effectively integrate the strengths of both models, combining the robust distance estimation capability of the SED-SDE model with the enhanced localization accuracy of the SED-DOA-OnePeace model, which benefits from contextual ONE-PEACE embeddings.

6. EXPERIMENTAL SETUP

The ReCoOP framework was trained using task-specific composite loss functions. For the SED-DOA-OnePeace model, binary cross-entropy loss was used for sound event detection, and mean squared error loss was applied to DOA estimation, gated by SED predictions to focus localization only on active events. For the SED-SDE model, binary cross-entropy loss was similarly used for detection, while the distance regression loss was masked by active event labels and stabilized to avoid numerical issues. In both cases, the SED and localization losses were weighted with a ratio of 0.1 to 1.

Training was conducted for 300 epochs with a batch size of 32, using the Tri-Stage Learning Rate Scheduler to ensure stable convergence. The best model checkpoints were selected based on macro F1-score performance on the dev-test split, which served as the validation set throughout training.

7. RESULTS AND DISCUSSION

As seen in Table 1, the progression from Experiment 1 to Experiment 4 demonstrates the impact of feature engineering, contextual modeling, and refinement strategies on stereo SELD performance. Experiment 1 uses the official baseline architecture [11] with only log mel energies and achieved an F-score of 22.78%, indicating limited spatial modeling. In contrast, Experiment 2 adopts the ResNet-Conformer framework, which ensembles two separate models - one for SED-DOA and another for SED-SDE - trained on all six input features, including ILD, IPD (sine and cosine), and GCC. This setup led to a substantial performance boost, achieving an F-score of 46.3%, thereby validating the importance of spatially informative features and task-specific model specialization.

Building on this, Experiment 3 introduced ONE-PEACE embeddings into the SED-DOA model, resulting in the best overall performance with an F-score of 48.2%. Experiment 4 further applied Audio Channel Swapping (ACS) as a data augmentation strategy during training, which maintained strong performance with an F-score of 47.5%. These results affirm that combining directional cues, contextual embeddings, and task-specific architectural branches within the ResNet-Conformer ensemble significantly improve performance over the baseline.

8. CONCLUSION

In this work, we presented our approach for Task 3 of the DCASE 2025 Challenge. We introduced a set of stereo spatial features

Experiment	F-score-20	DOAE-CD (deg)	RDE-CD
Baseline (Log-Mel Only, Baseline Architecture) (1)	22.78%	24.5	0.41
ResNet-Conformer (All Features: 64 Mel + ILD/IPD/GCC) (2)	46.3%	13.0	0.37
ReCoOP: ResNet-Conformer + ONE-PEACE Embeddings (3)	48.2%	13.3	0.36
ACS (Audio Channel Swapping Augmentation) + ReCoOP (4)	47.5%	13.7	0.36

Table 1: Comparison of SELD performance across different experimental configurations - All results have been reported on the dev-test split of the DCASE2025 Task 3 Stereo SELD Dataset. All experiments employ synthetic audio data augmentation, while Audio Channel Swapping (ACS) augmentation is used exclusively in Experiment 4. Experiment 3 refers to our submission system 1, while Experiment 4 corresponds to our submission system 2.

as input to our proposed ReCoOP framework, which builds on a backbone of ResNet and Conformer modules. While the ResNet-Conformer architecture follows a similar structure to last year's top-performing models, we extended it by integrating the pre-trained ONE-PEACE model, forming an ensemble with the ResNet-Conformer module. This ensemble was further fine-tuned using two additional Conformer layers to enhance downstream performance.

Our experimental results demonstrate that integrating ONE-PEACE embeddings into the ResNet-Conformer architecture yields an additional 2% improvement in performance compared to using the ResNet-Conformer module alone. Compared to the official DCASE baseline, our method achieves an overall 25.42% performance gain. These results highlight the benefit of combining pre-trained models with task-specific architectures and fine-tuning them jointly. Additionally, the introduction of stereo spatial features, alongside conventional Mel spectrogram inputs, proves valuable in improving SELD performance under stereo audio conditions.

Future directions can include refining the fusion of ONE-PEACE embeddings with audio features through improved temporal alignment. We also aim to evaluate generalization on other stereo SELD datasets and explore self-supervised learning for enhanced spatial feature extraction.

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